

Fig. 5: Component layout for the PCB

subroutine. When pressing of the switch is detected, 'count' variable is incremented continuously in the loop and as soon as the key is released, the counter is stopped and 'dice_count' is computed; which is then used as a dice number and displayed using 'disp_Char' subroutine. The font map of the dice number is listed in 'font5x7.h' file. In this file, the data is listed out in array form and display values are picked up from the array using the offset method.

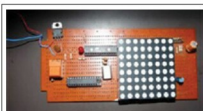
When the switch is released, an animated pattern appears on the display, which indicates that the display is waiting for a roll. The animated data is also kept in the above-mentioned font file (at offset value of 131 and 130 location). Subroutine

TABLE II
Program Files Used in the Software

File name	Functionality
Dice.c	Check the status of the pushbutton and compute the dice number
Dice.h	Declare global functions and constant values
MAX7219.c	Control the operation of MAX7219 (generation of serial clock, conversion of 8-bit address and data into 16-bit serial data, etc)
makefile	Contains target microcontroller and project declarations (definitions) along with compiling and debugging options, etc
Font5x7.h	Contains the character map for displaying the information

TABLE III
Subroutines Used in the Max7219.c Program

Subroutine name	Functionality
init_7219 (void)	Initialise MAX7219 and display. Set the scan limit, configure MAX7219 in 'No Decode' mode and also set the intensity of the LED matrix
display_Matrix()	Actual display of information on the LED matrix
shut_Down_7219()	Put MAX7219 in shutdown mode
display_Test_7219	Test the LED matrix
display_Norm_7219()	Display information (normal-mode operation)
clear_7219()	Clear the display
disp_Char(char character)	Read the font data for displaying the information
delay(char count)	Generate a delay
clear_Matrix()	Fill matrix[] with 0's



Assembled board



First animated screen pattern after the circuit is switched on



Second animated screen pattern



Dice-roll number '1'



Dice-roll number '6'



Dice-roll number '3'

Fig. 6: Project in various stages of operation